

```
> with(student); np := 8:
[D, Diff, Doubleint, Int, Limit, Lineint, Product, Sum, Tripleint, changevar, completesquare,
distance, equate, integrand, intercept, intparts, leftbox, leftsum, makeproc, middlebox, middlesum,
midpoint, powsubs, rightbox, rightsum, showtangent, simpson, slope, summand, trapezoid]
```

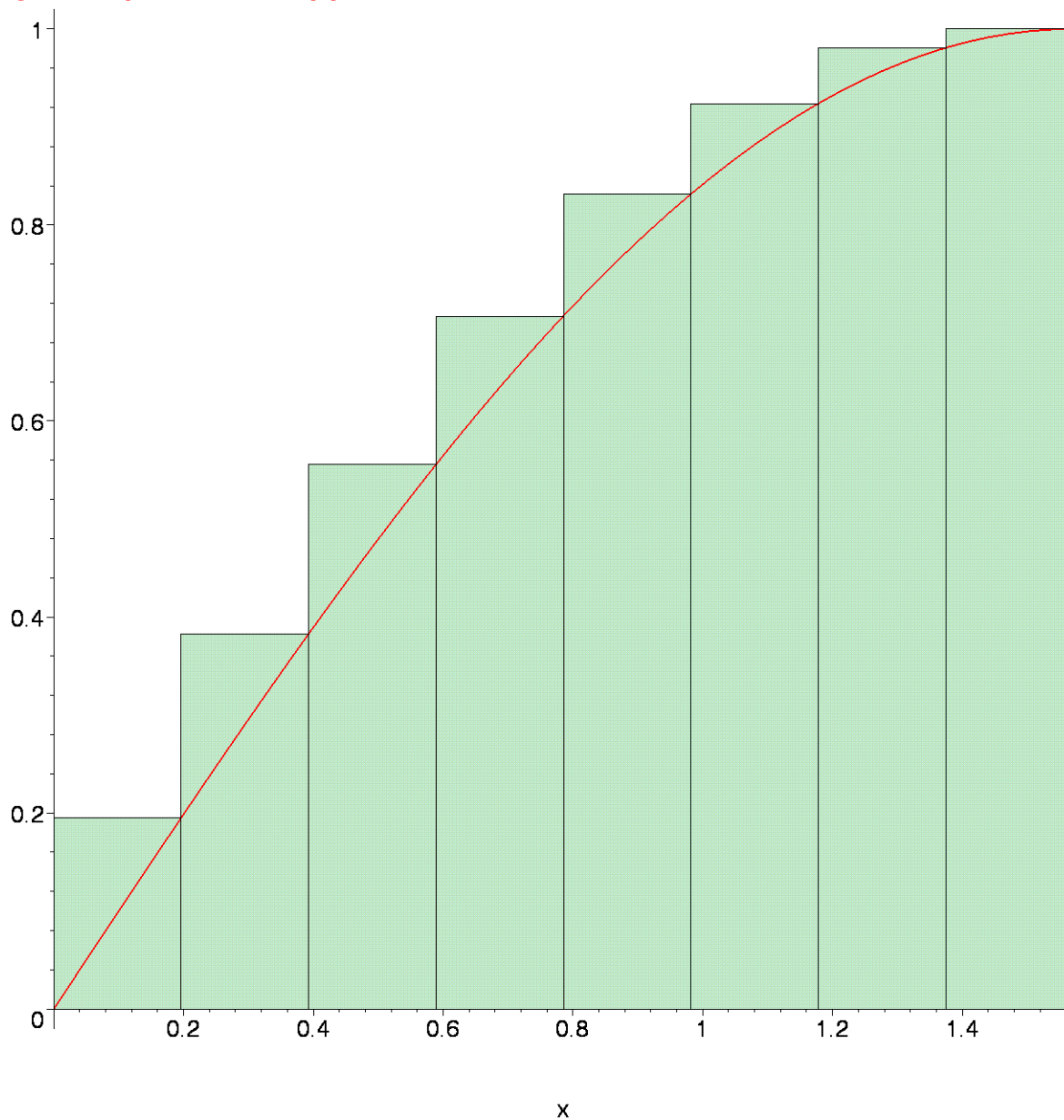
```
> f := sin(x); a := 0; b := Pi/2;
```

$f := \sin(x)$

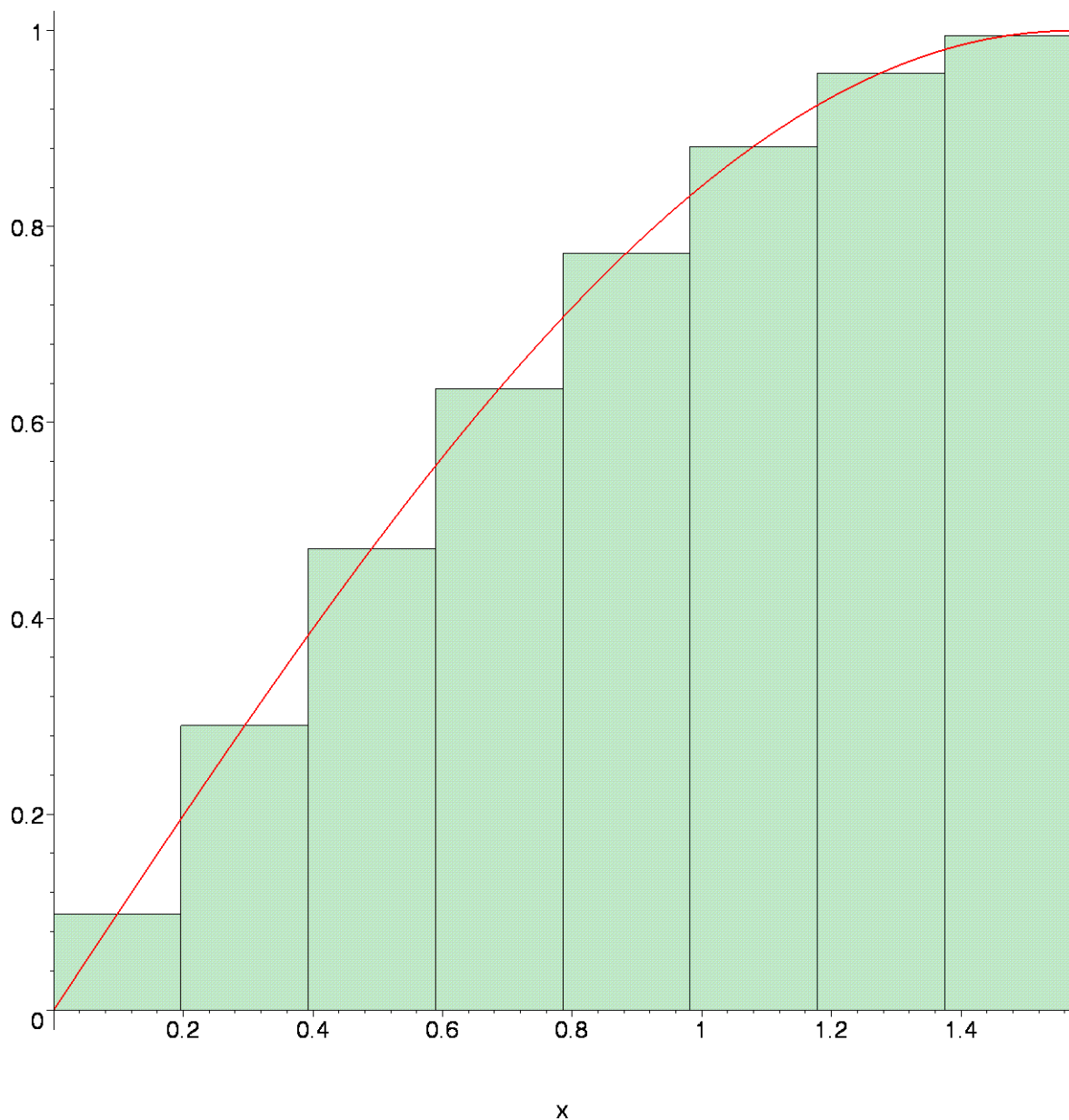
$a := 0$

$b := \frac{\pi}{2}$

```
> rightbox(f, x=a..b, np);
```



```
> middlebox(f, x=a..b, np);
```



```

> n := 8;
  r := evalf(rightsum(f,x=a..b,n));
  m := evalf(middlesum(f,x=a..b,n));
  t := evalf(trapezoid(f,x=a..b,n));
  s := evalf(simpson(f,x=a..b,n));

      n := 8
      r := 1.094959942
      m := 1.001608189
      t := 0.9967851722
      s := 1.000008295

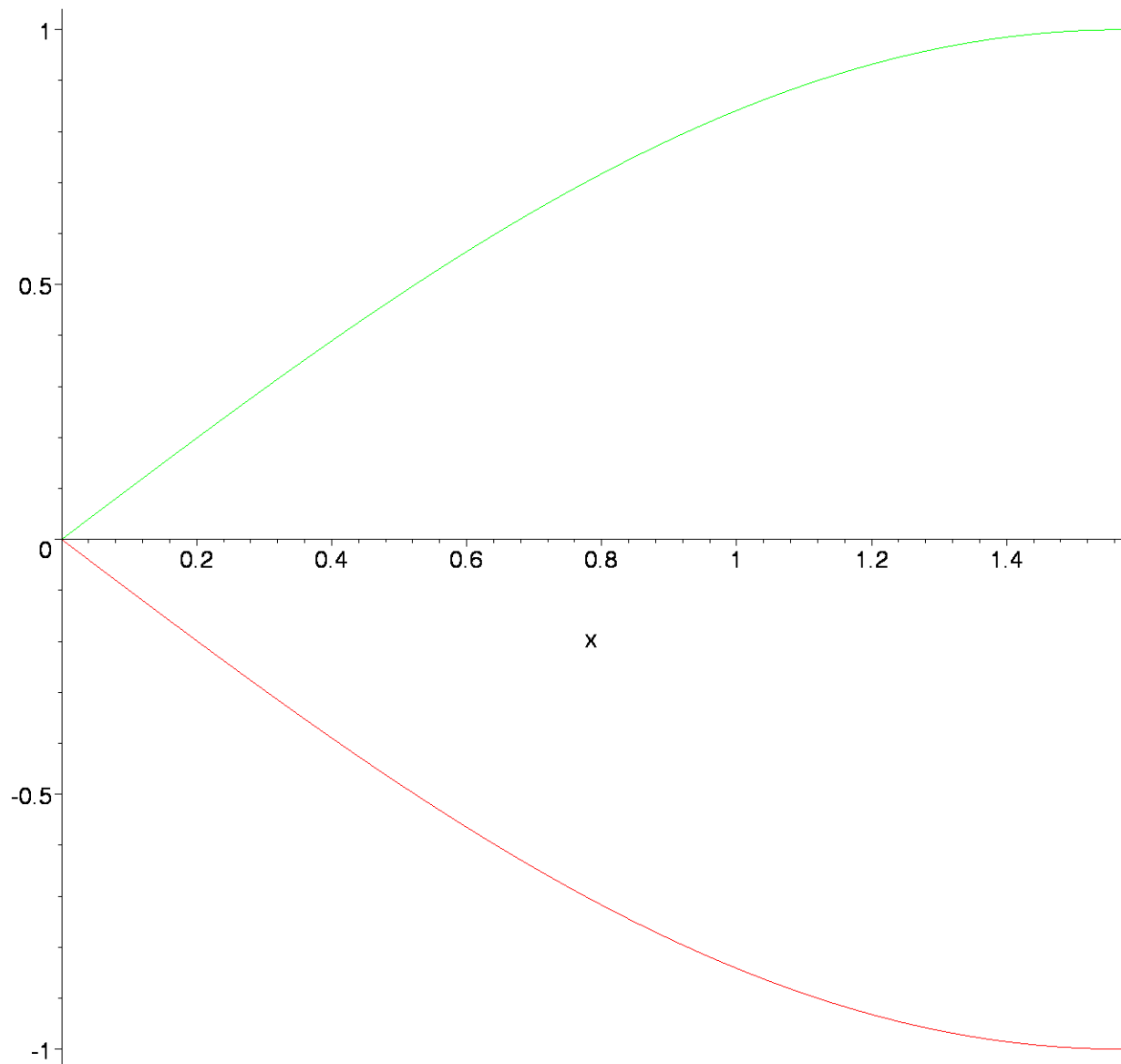
> fpp := diff(f,x$2);
  fpppp := diff(f,x$4);

```

```
plot([fpp, fpppp], x=a..b, color=[red, green]);
```

```
fpp := -sin(x)
```

```
fpppp := sin(x)
```



```
> M := 1; K := 1;
err_mid := evalf((b-a)^3*(M)/(24*n^2));
err_trap := evalf((b-a)^3*(M)/(12*n^2));
err_simp := evalf((b-a)^5*(K)/(180*n^4));

M := 1
K := 1
err_mid := 0.002523297257
err_trap := 0.005046594514
err_simp := 0.00001297080506
```

```
> evalf(int(f,x=a..b));
```

1.

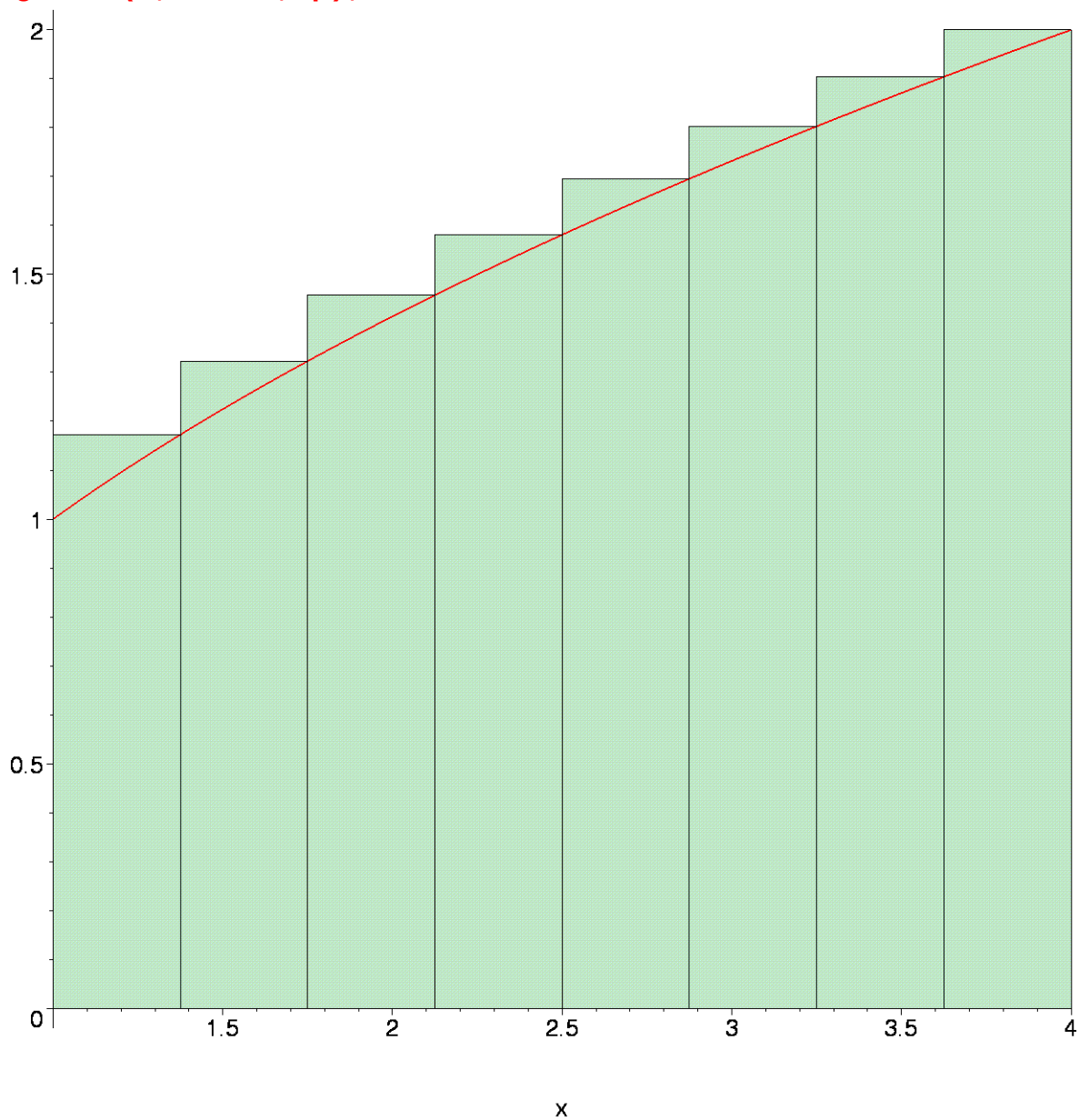
```
> f := sqrt(x); a := 1; b := 4;
```

$$f := \sqrt{x}$$

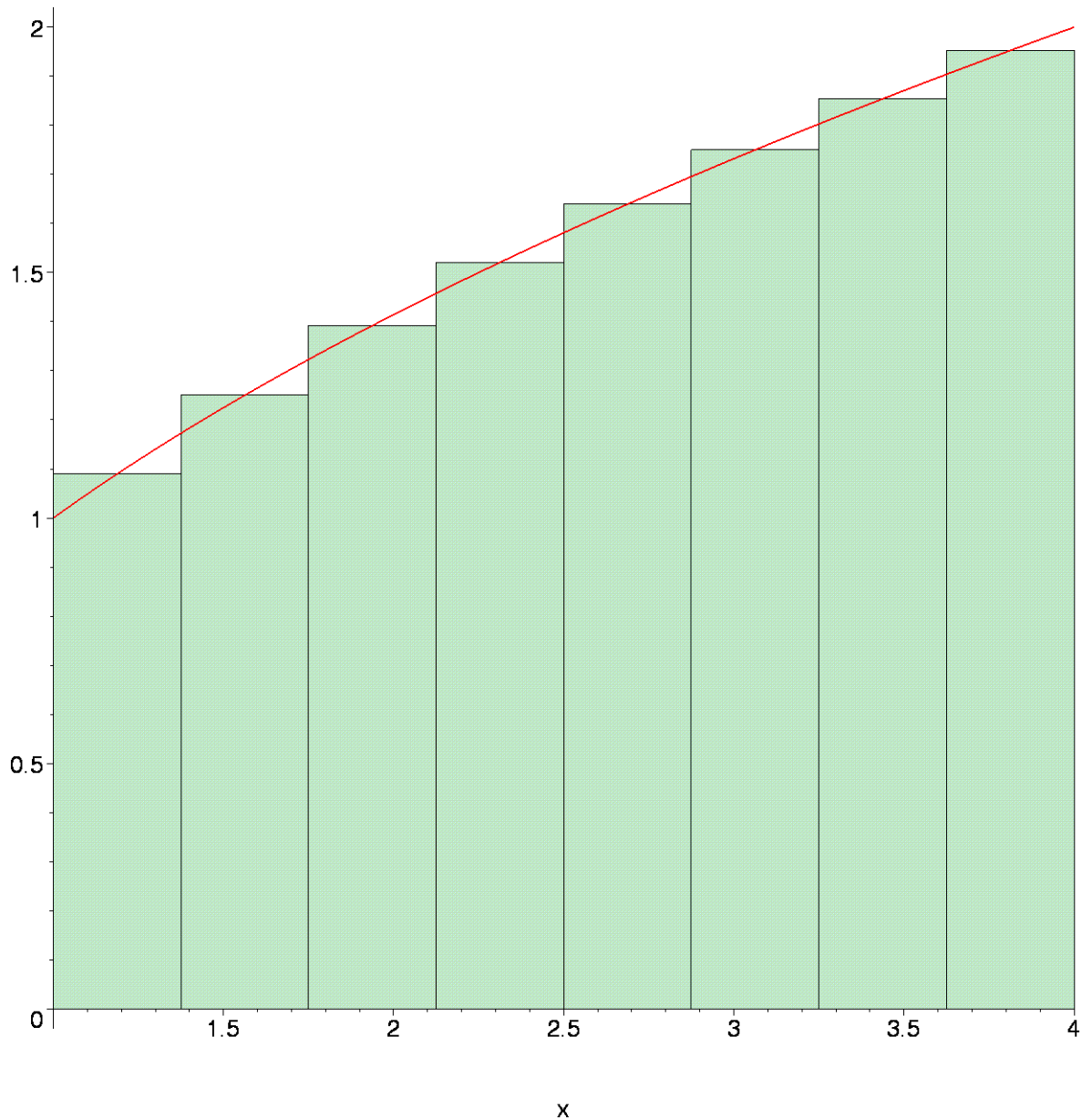
$$a := 1$$

$$b := 4$$

```
> rightbox(f,x=a..b,np);
```



```
> middlebox(f,x=a..b,np);
```



```

> n := 8;
  r := evalf(rightsum(f,x=a..b,n));
  m := evalf(middlesum(f,x=a..b,n));
  t := evalf(trapezoid(f,x=a..b,n));
  s := evalf(simpson(f,x=a..b,n));

                                n := 8
                                r := 4.851246679
                                m := 4.668123049
                                t := 4.663746679
                                s := 4.666631374

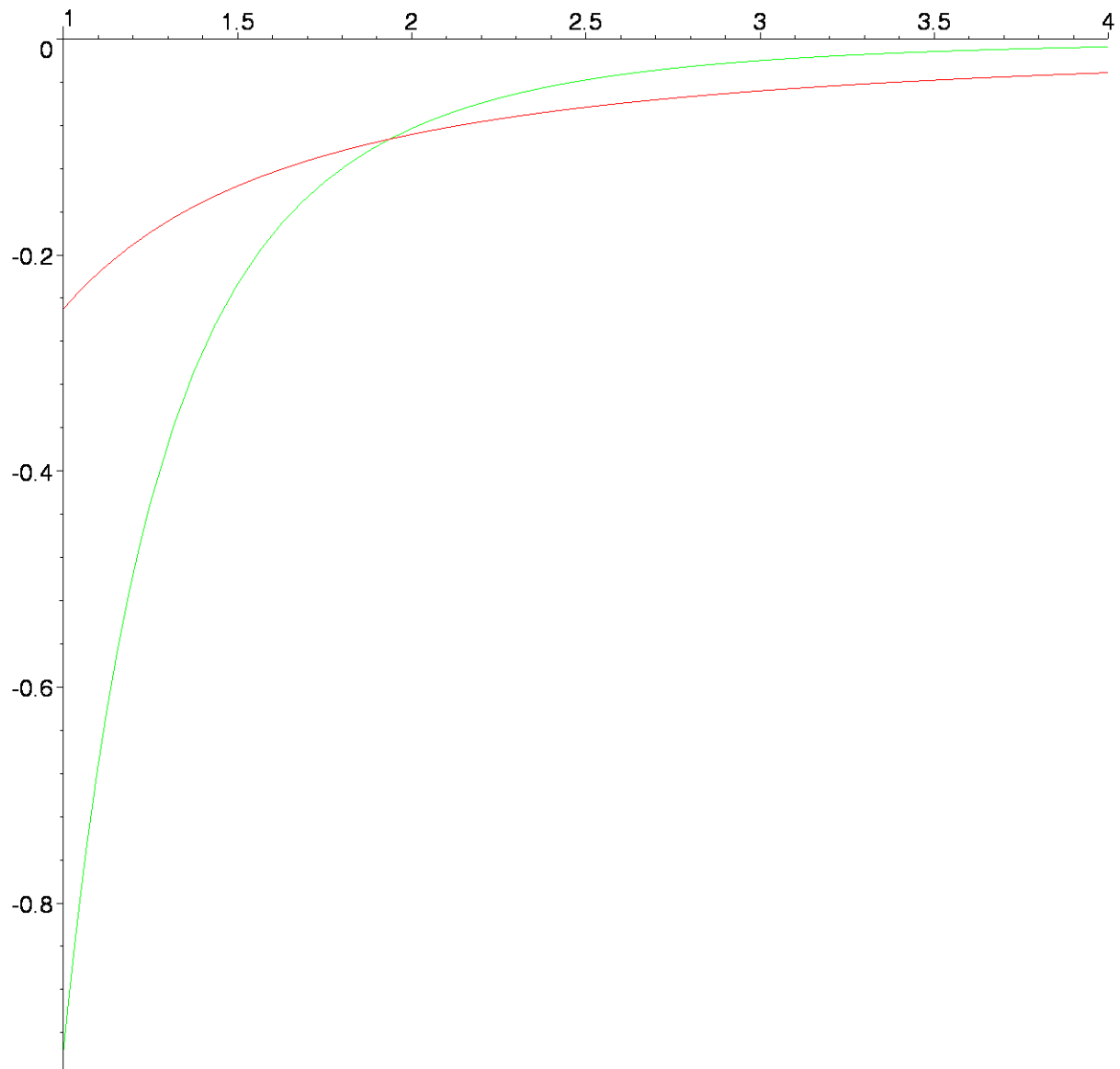
> fpp := diff(f,x$2);
  fpppp := diff(f,x$4);

```

```
plot([fpp, fpppp], x=a..b, color=[red, green]);
```

$$fpp := -\frac{1}{4x^{(3/2)}}$$

$$fpppp := -\frac{15}{16x^{(7/2)}}$$



```
> M := 1/4; K := 1;  
err_mid := evalf((b-a)^3*(M)/(24*n^2));  
err_trap := evalf((b-a)^3*(M)/(12*n^2));  
err_simp := evalf((b-a)^5*(K)/(180*n^4));
```

$$M := \frac{1}{4}$$

```
K := 1  
err_mid := 0.004394531250  
err_trap := 0.008789062500  
err_simp := 0.0003295898438
```

```
> evalf(int(f,x=a..b));
```

```
4.666666667
```

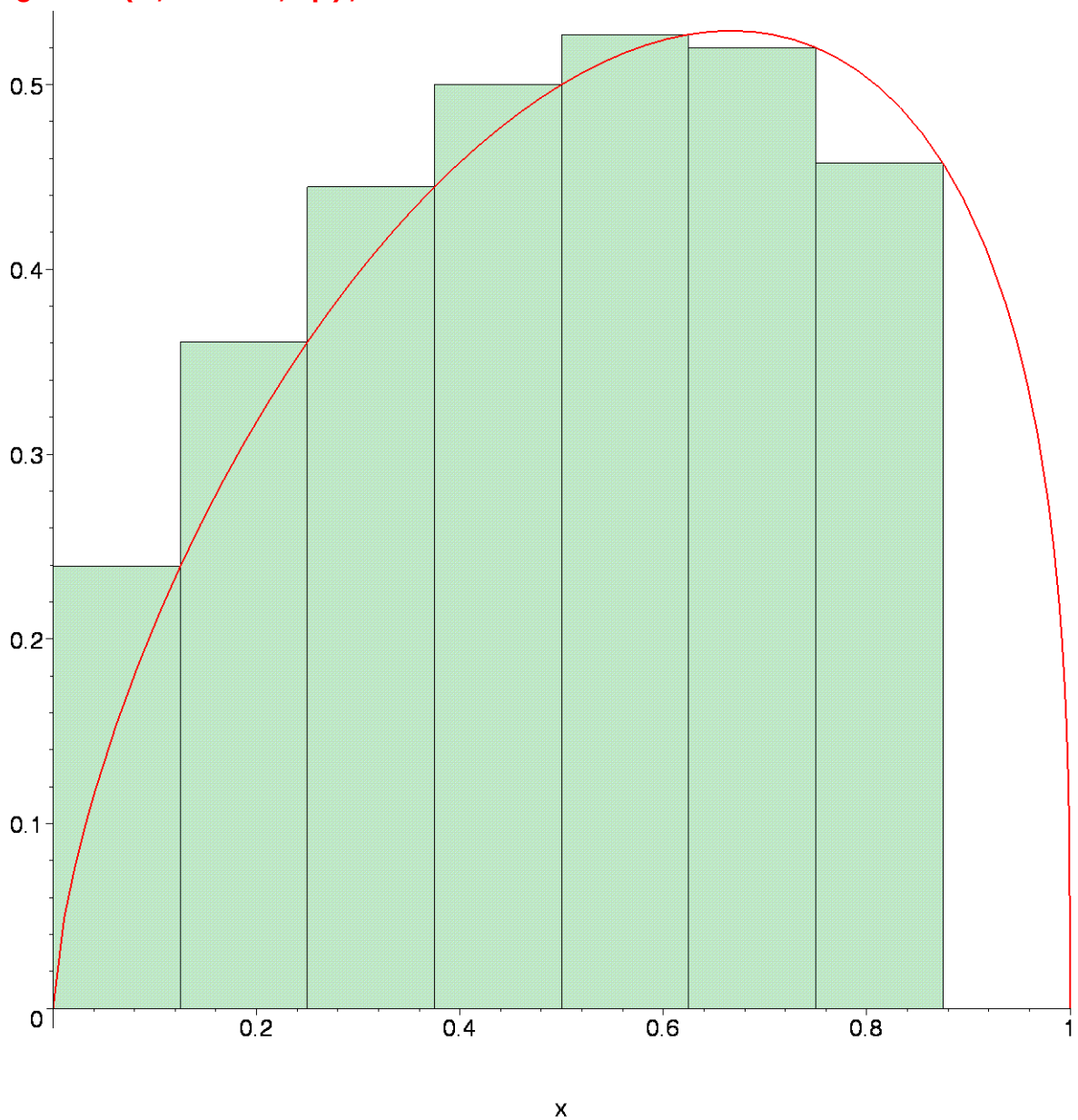
```
> f := x^(2/3)*(1-x)^(1/3); a := 0; b := 1;
```

```
 $f := x^{(2/3)}(1-x)^{(1/3)}$ 
```

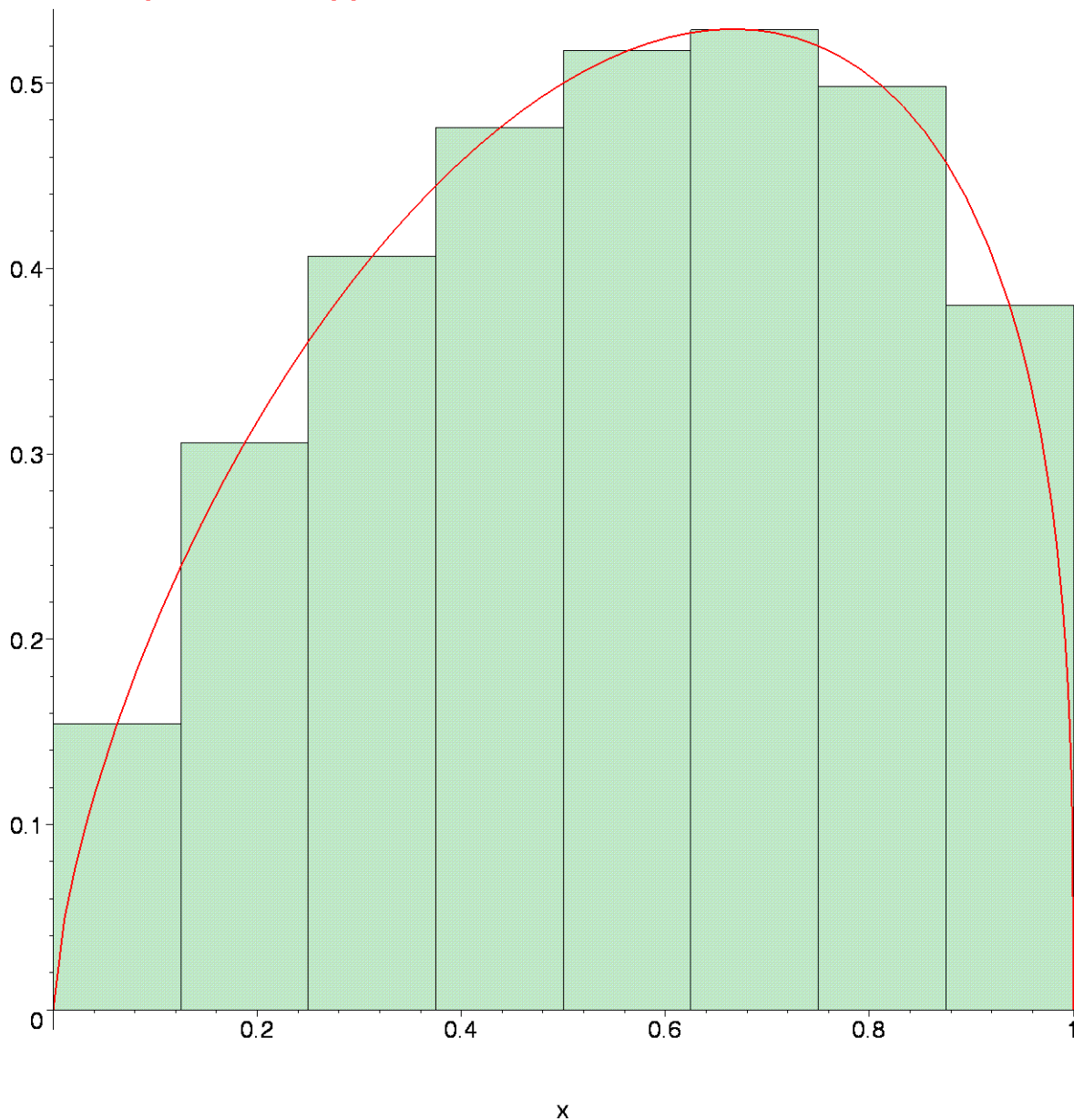
```
a := 0
```

```
b := 1
```

```
> rightbox(f,x=a..b,np);
```



```
> middlebox(f,x=a..b,np);
```



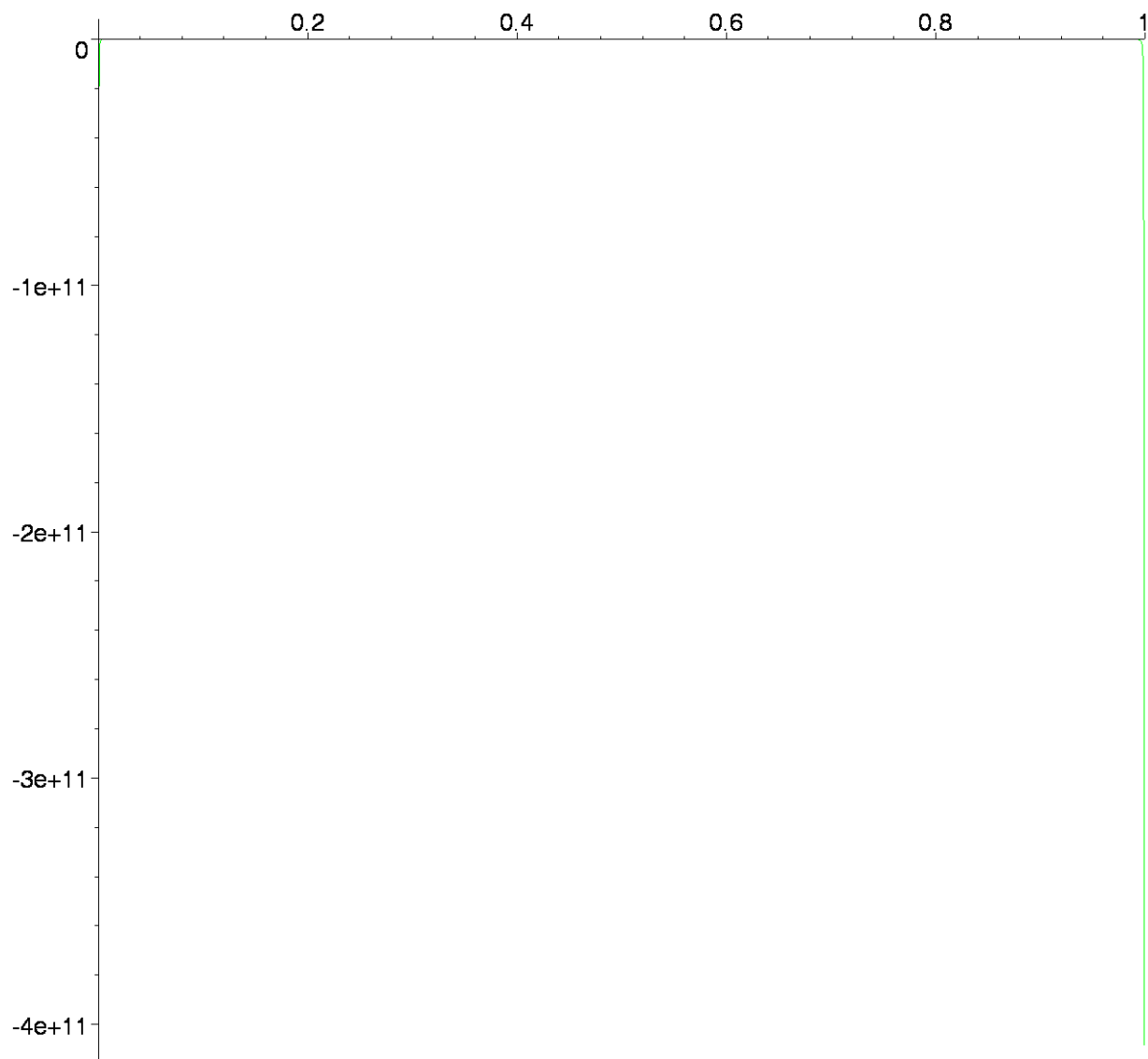
```
> n := 8;  
r := evalf(rightsum(f,x=a..b,n));  
m := evalf(middlesum(f,x=a..b,n));  
t := evalf(trapezoid(f,x=a..b,n));  
s := evalf(simpson(f,x=a..b,n));  
  
n := 8  
r := 0.3811087549  
m := 0.4082995574  
t := 0.3811087549  
s := 0.3930963940  
  
> fpp := diff(f,x$2);
```

```
fpppp := diff(f,x$4);
plot([fpp,fpppp],x=a..b,color=[red,green]);
```

$$fpp := -\frac{2(1-x)^{(1/3)}}{9x^{(4/3)}} - \frac{4}{9x^{(1/3)}(1-x)^{(2/3)}} - \frac{2x^{(2/3)}}{9(1-x)^{(5/3)}}$$

$$fpppp := -\frac{56(1-x)^{(1/3)}}{81x^{(10/3)}} - \frac{32}{81x^{(7/3)}(1-x)^{(2/3)}} + \frac{8}{27x^{(4/3)}(1-x)^{(5/3)}} \\ - \frac{80}{81x^{(1/3)}(1-x)^{(8/3)}} - \frac{80x^{(2/3)}}{81(1-x)^{(11/3)}}$$

x



```
> M := infinity; K := infinity;
```

$M := \infty$

$K := \infty$

```
> evalf(int(f,x=a..b));
```

0.4030665255

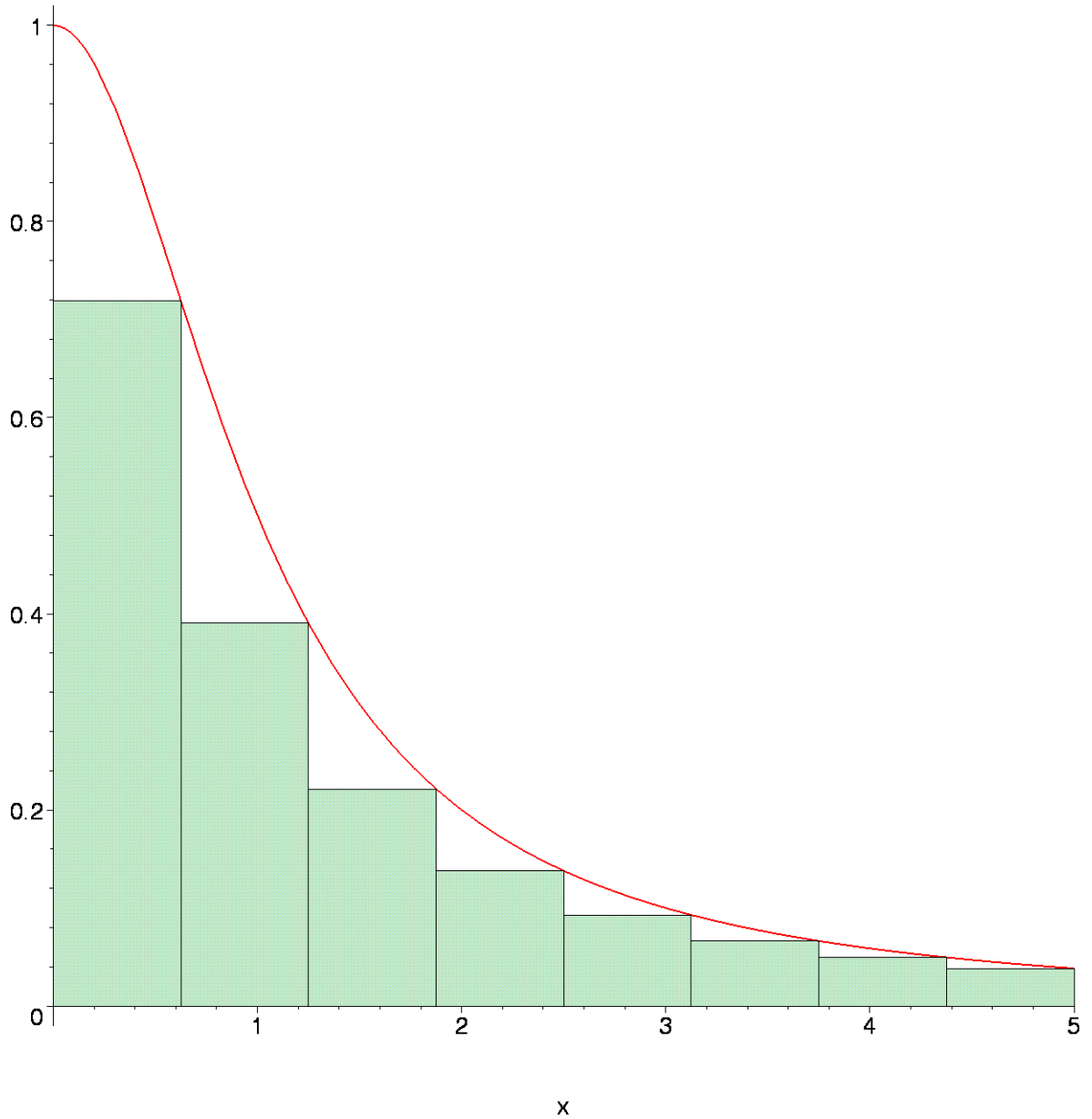
```
> f := 1/(x^2+1); a := 0; b := 5;
```

$$f := \frac{1}{x^2 + 1}$$

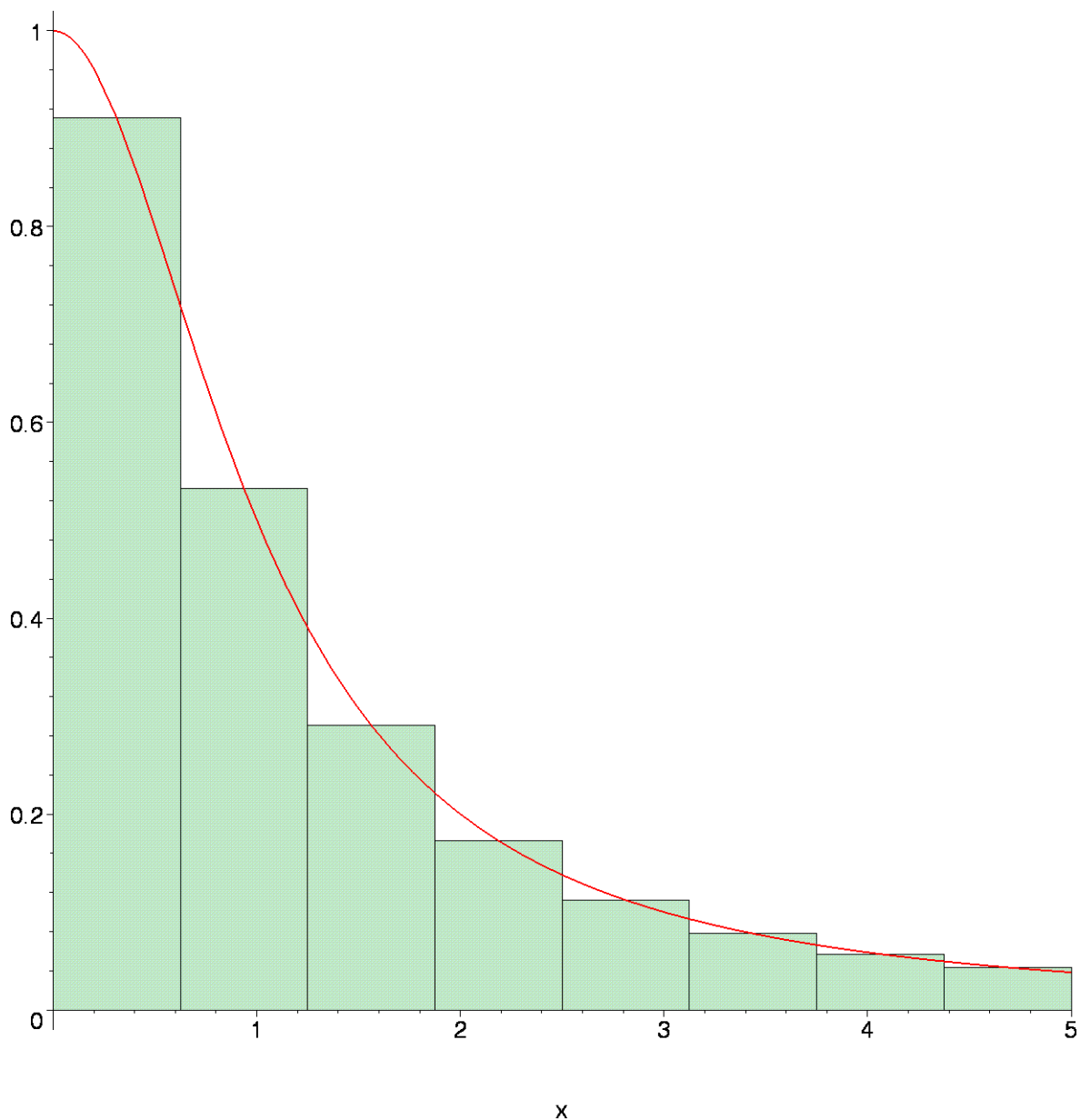
$a := 0$

$b := 5$

```
> rightbox(f,x=a..b,np);
```



```
> middlebox(f,x=a..b,np);
```



```

> n := 8;
  r := evalf(rightsum(f,x=a..b,n));
  m := evalf(middlesum(f,x=a..b,n));
  t := evalf(trapezoid(f,x=a..b,n));
  s := evalf(simpson(f,x=a..b,n));

                                n := 8
                                r := 1.072575040
                                m := 1.373505133
                                t := 1.373055809
                                s := 1.366659518

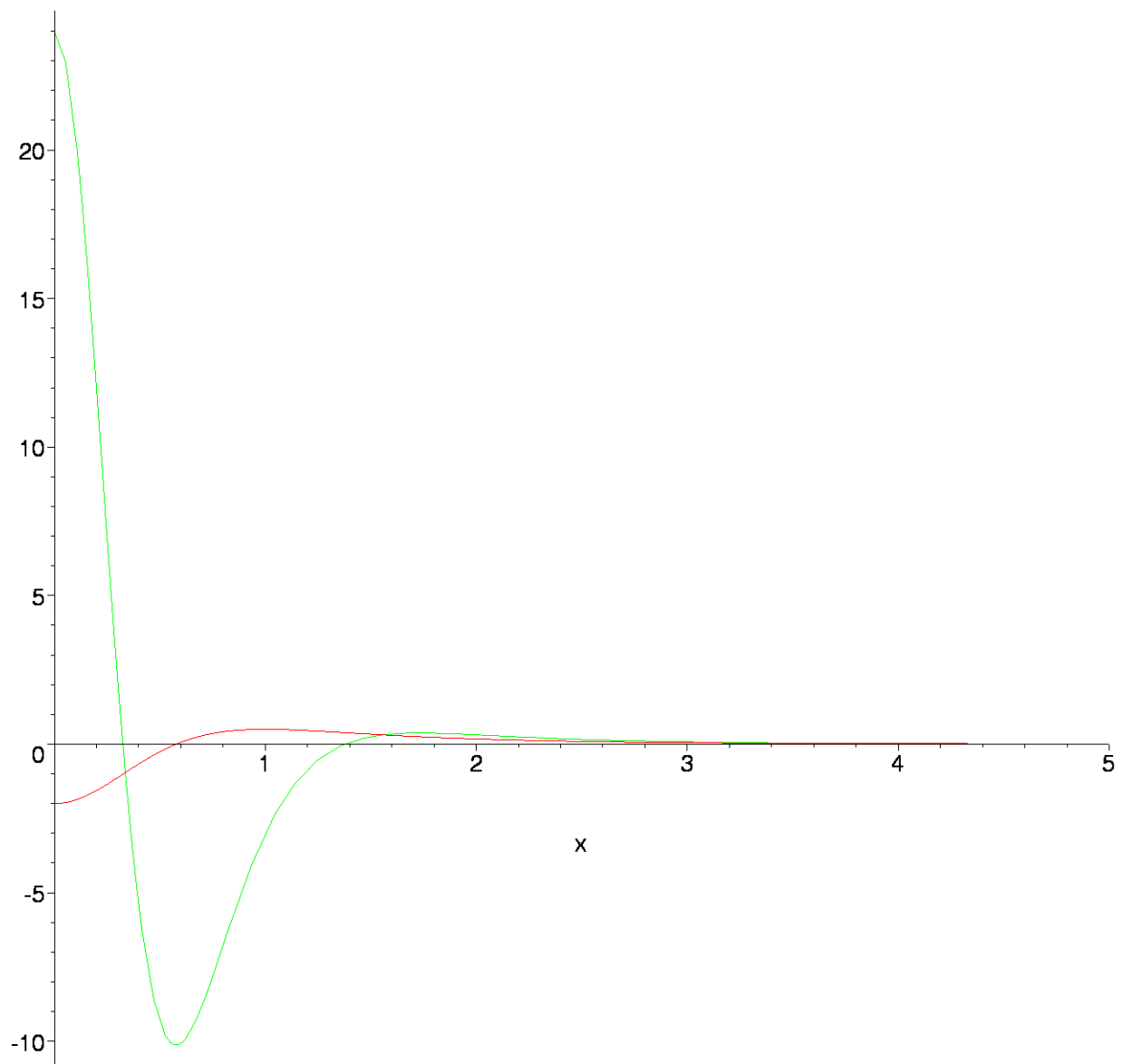
> fpp := normal(diff(f,x$2));
  fpppp := normal(diff(f,x$4));

```

```
plot([fpp, fpppp], x=a..b, color=[red, green]);
```

$$fpp := \frac{2(3x^2 - 1)}{(x^2 + 1)^3}$$

$$fpppp := \frac{24(5x^4 - 10x^2 + 1)}{(x^2 + 1)^5}$$



```
> M := 2; K := 24;
err_mid := evalf((b-a)^3*(M)/(24*n^2));
err_trap := evalf((b-a)^3*(M)/(12*n^2));
err_simp := evalf((b-a)^5*(K)/(180*n^4));
```

$M := 2$

$K := 24$

```
err_mid := 0.1627604167  
err_trap := 0.3255208333  
err_simp := 0.1017252604
```

```
> evalf(int(f,x=a..b));
```

```
1.373400767
```

```
> nn := 200;
```

```
err_mid := evalf((b-a)^3*(M)/(24*nn^2));  
err_trap := evalf((b-a)^3*(M)/(12*nn^2));  
err_simp := evalf((b-a)^5*(K)/(180*nn^4));
```

```
nn := 200
```

```
err_mid := 0.0002604166667
```

```
err_trap := 0.0005208333333
```

```
err_simp := 0.2604166667 10-6
```

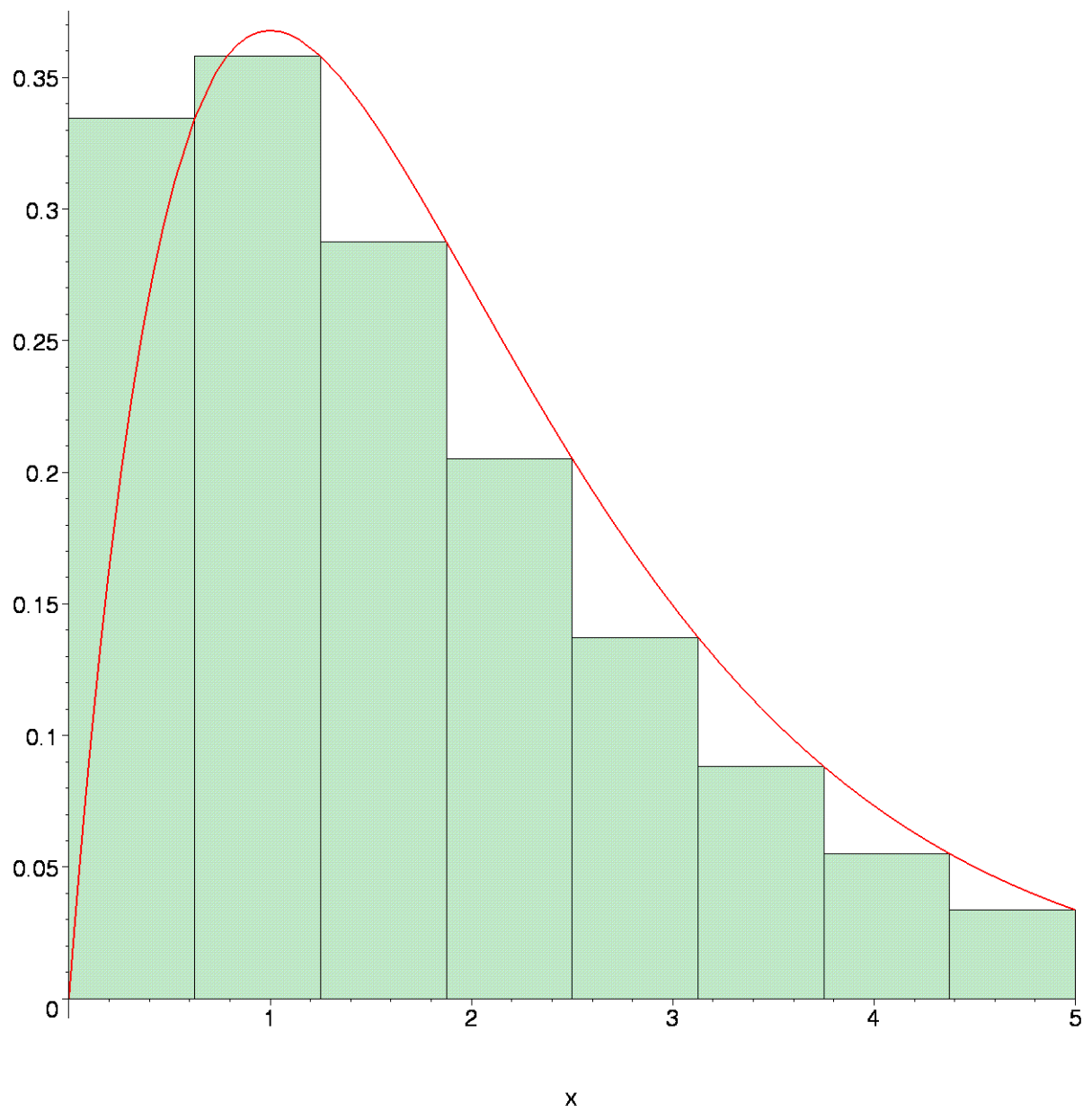
```
> f := x*exp(-x); a := 0; b := 5;
```

```
 $f := x e^{(-x)}$ 
```

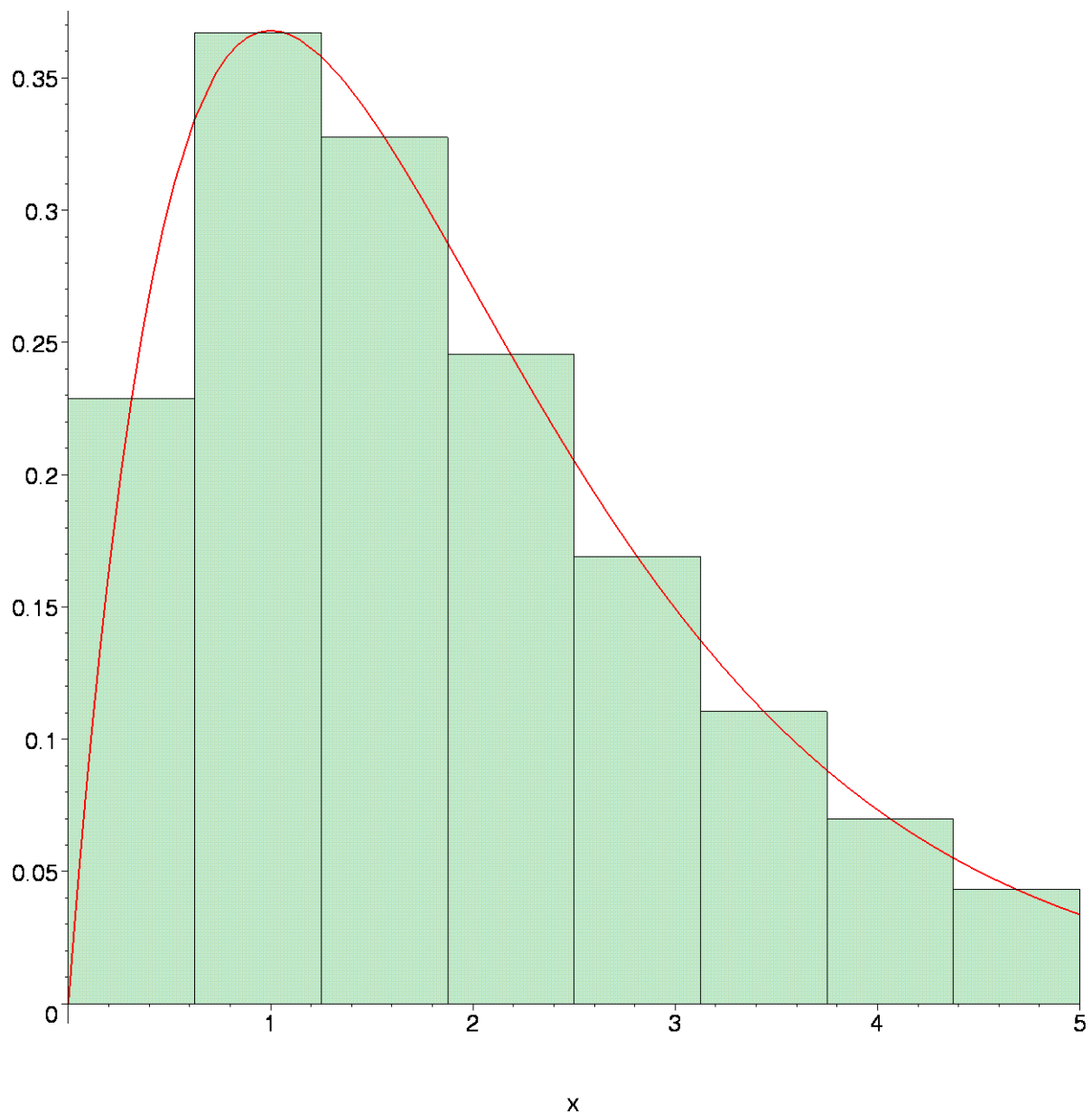
```
a := 0
```

```
b := 5
```

```
> rightbox(f,x=a..b,np);
```



```
> middlebox(f,x=a..b,np);
```



```

> n := 8;
  r := evalf(rightsum(f,x=a..b,n));
  m := evalf(middlesum(f,x=a..b,n));
  t := evalf(trapezoid(f,x=a..b,n));
  s := evalf(simpson(f,x=a..b,n));

                                n := 8
                                r := 0.9372998562
                                m := 0.9757376350
                                t := 0.9267718141
                                s := 0.9572041242

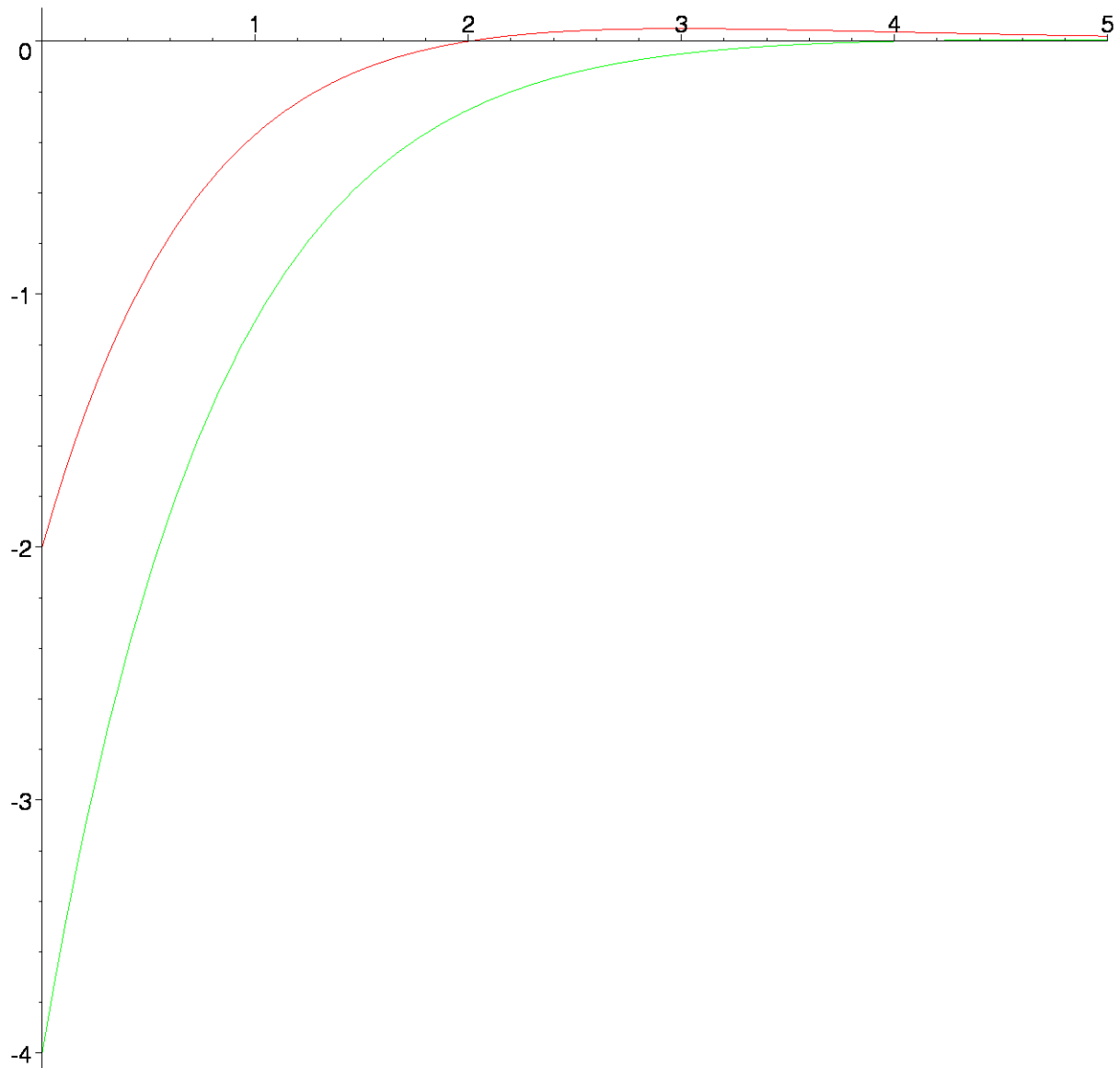
> fpp := diff(f,x$2);
  fpppp := diff(f,x$4);

```

```
plot([fpp, fpppp], x=a..b, color=[red, green]);
```

$$fpp := -2 e^{(-x)} + x e^{(-x)}$$

$$fpppp := -4 e^{(-x)} + x e^{(-x)}$$



```
> M := 2; K := 4;
```

```
err_mid := evalf((b-a)^3*(M)/(24*n^2));
```

```
err_trap := evalf((b-a)^3*(M)/(12*n^2));
```

```
err_simp := evalf((b-a)^5*(K)/(180*n^4));
```

```
M := 2
```

```
K := 4
```

```
err_mid := 0.1627604167
```

```
err_trap := 0.3255208333
```

$err_{simp} := 0.01695421007$

```
> evalf(int(f,x=a..b));
```

0.9595723180

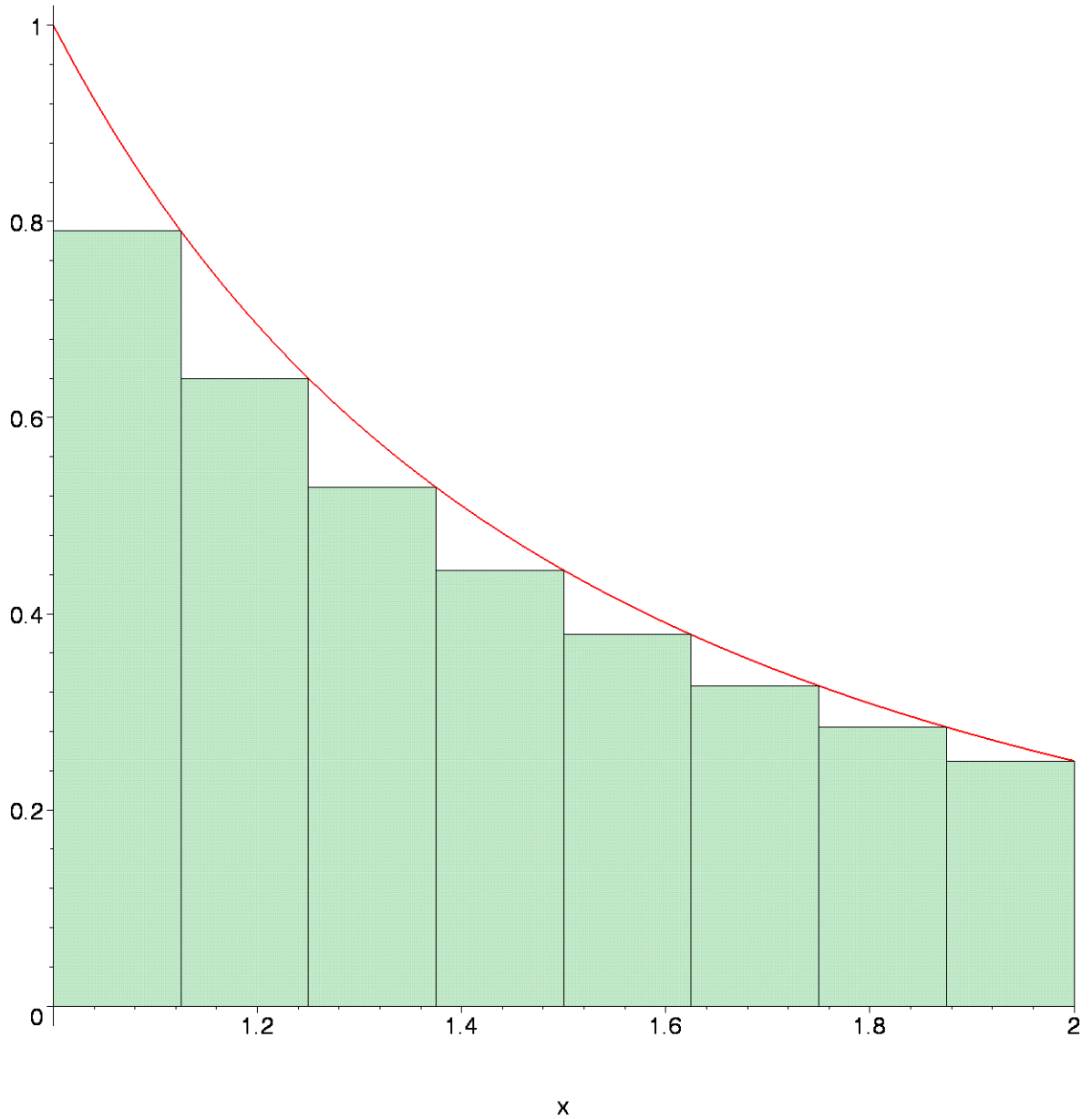
```
> f := 1/x^2;a := 1; b := 2;
```

$$f := \frac{1}{x^2}$$

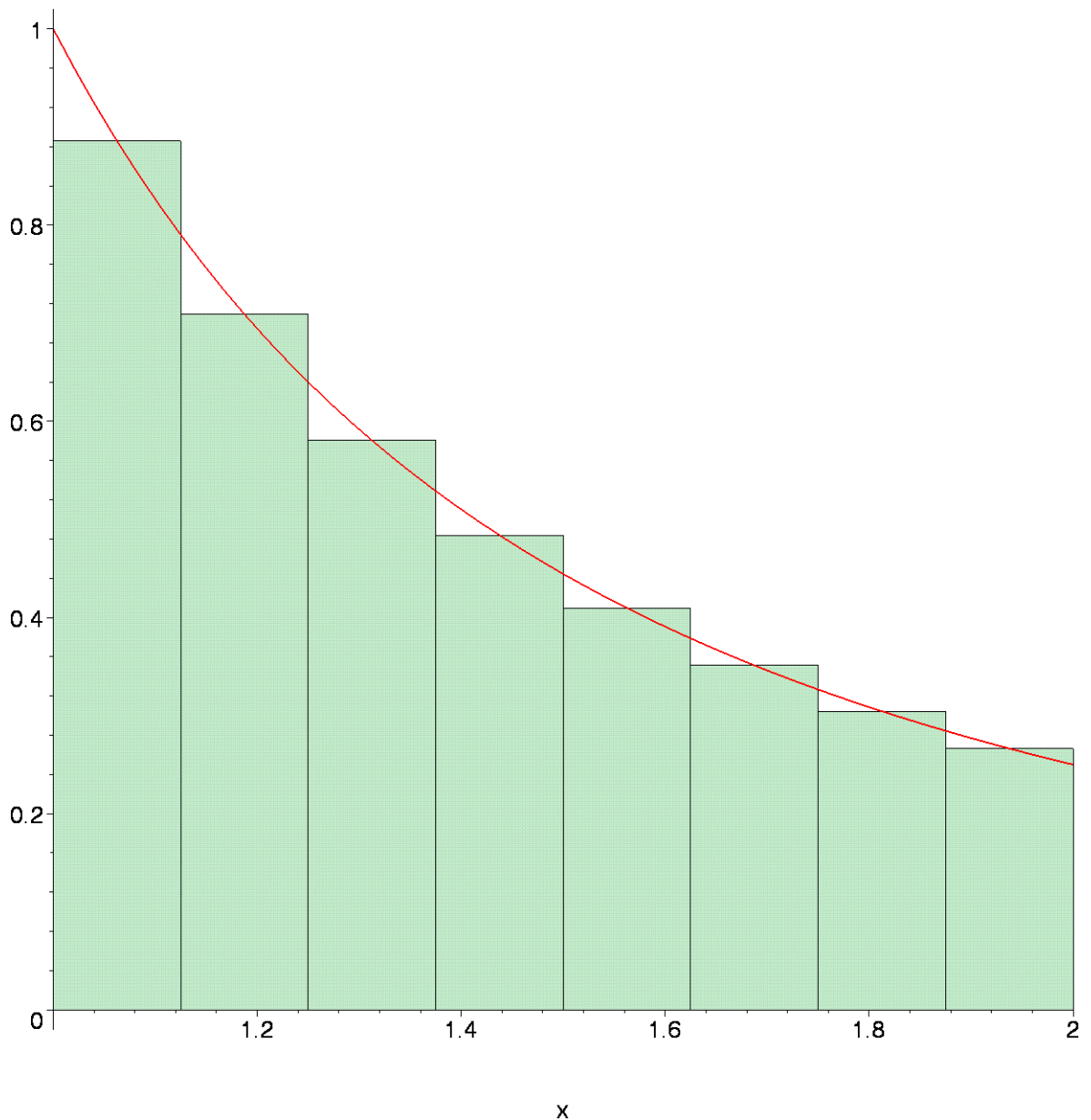
$a := 1$

$b := 2$

```
> rightbox(f,x=a..b,np);
```



```
> middlebox(f,x=a..b,np);
```



```

> n := 8;
  r := evalf(rightsum(f,x=a..b,n));
  m := evalf(middlesum(f,x=a..b,n));
  t := evalf(trapezoid(f,x=a..b,n));
  s := evalf(simpson(f,x=a..b,n));

                                n := 8
                                r := 0.4553958502
                                m := 0.4988674899
                                t := 0.5022708502
                                s := 0.5000298791

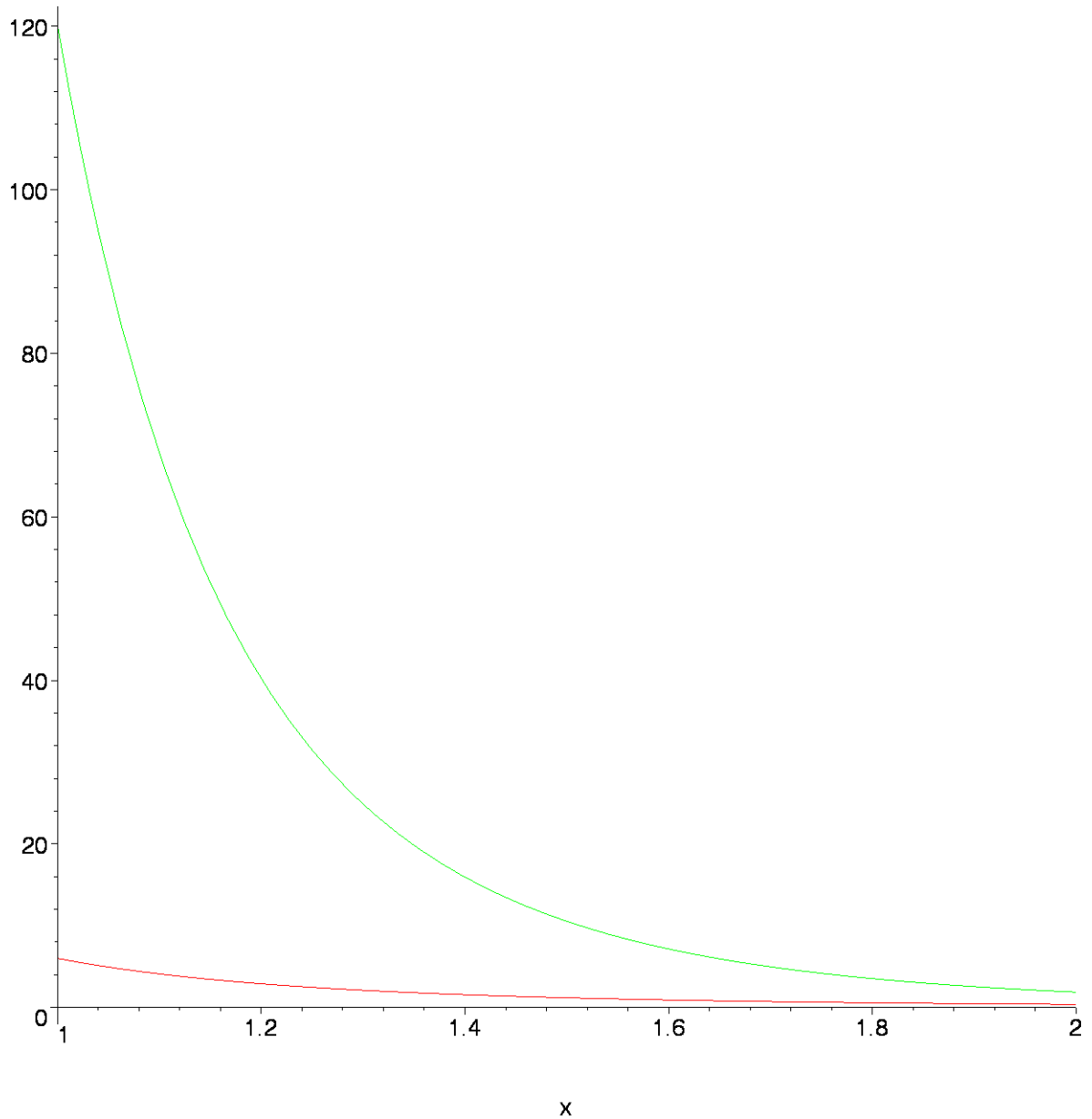
> fpp := diff(f,x$2);
  fpppp := diff(f,x$4);

```

```
plot([fpp, fpppp], x=a..b, color=[red, green]);
```

$$fpp := \frac{6}{x^4}$$

$$fpppp := \frac{120}{x^6}$$



```
> M := 6; K := 120;  
err_mid := evalf((b-a)^3*(M)/(24*n^2));  
err_trap := evalf((b-a)^3*(M)/(12*n^2));  
err_simp := evalf((b-a)^5*(K)/(180*n^4));  
  
M:= 6  
K:= 120  
err_mid := 0.003906250000
```

```
err_trap := 0.007812500000  
err_simp := 0.0001627604167
```

```
> evalf(int(f,x=a..b));
```

```
0.50000000000
```

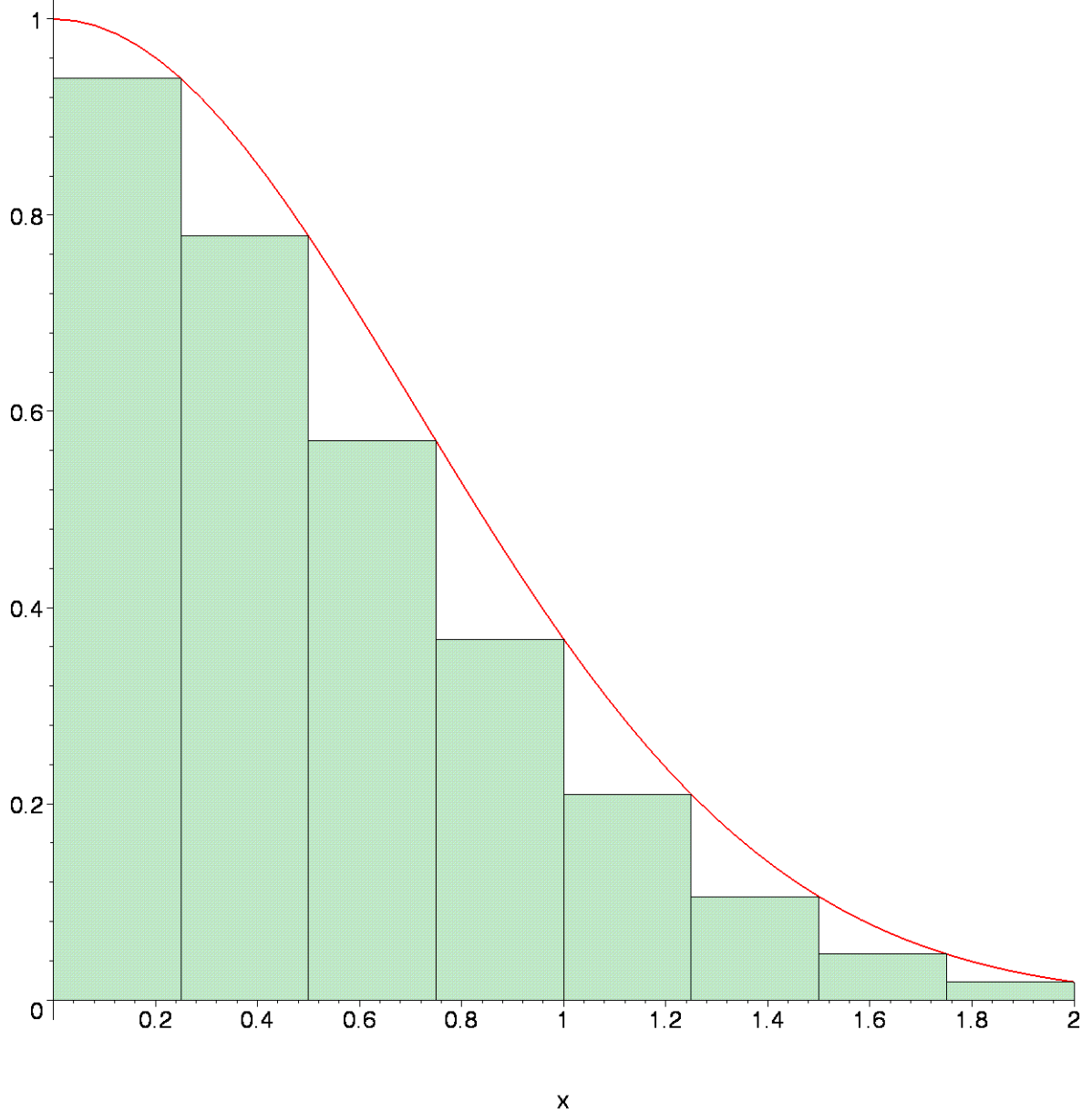
```
> f := exp(-x^2); a := 0; b := 2;
```

$f := e^{(-x^2)}$

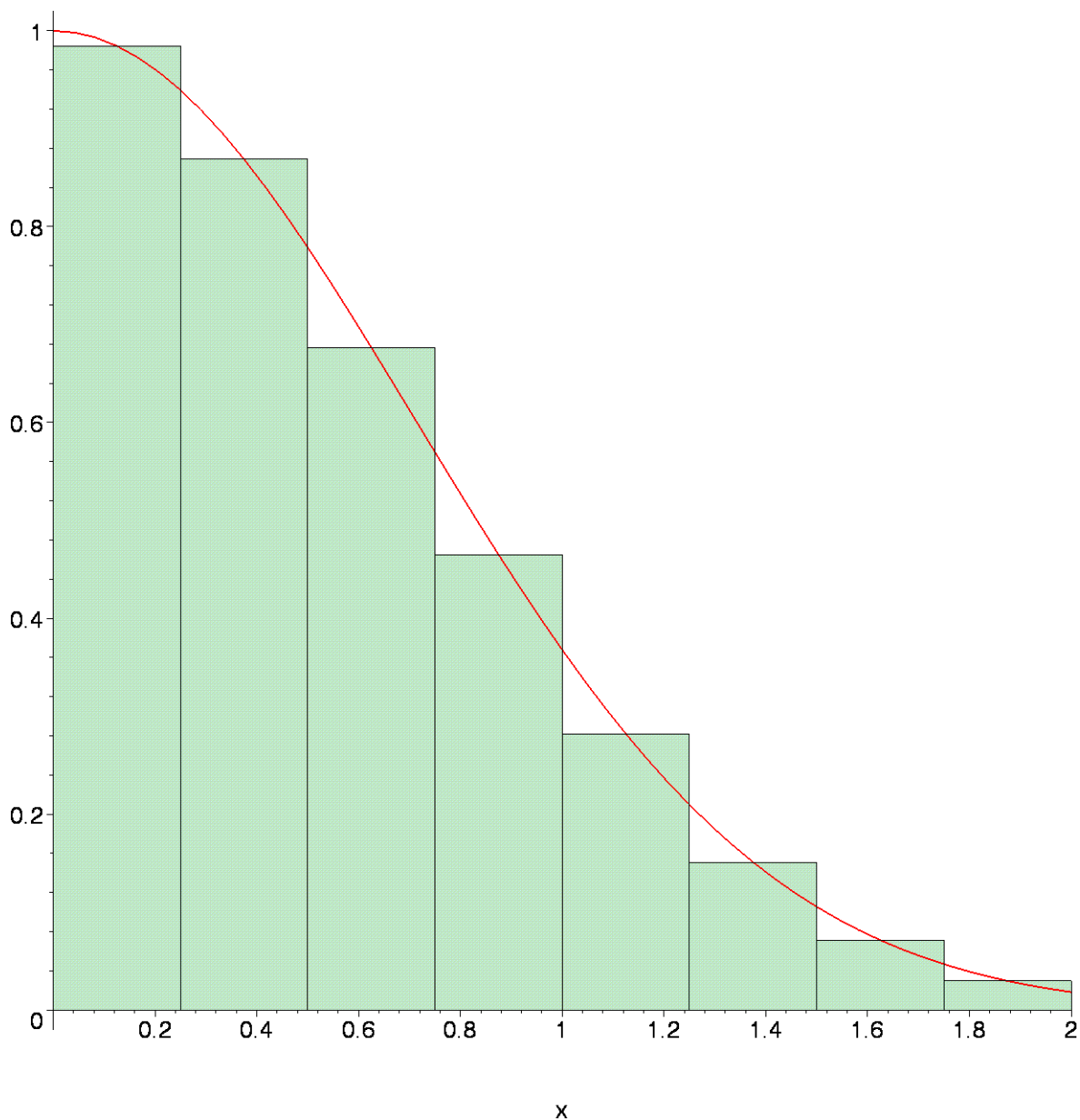
$a := 0$

$b := 2$

```
> rightbox(f,x=a..b,np);
```



```
> middlebox(f,x=a..b,np);
```



```

> n := 8;
  r := evalf(rightsum(f,x=a..b,n));
  m := evalf(middlesum(f,x=a..b,n));
  t := evalf(trapezoid(f,x=a..b,n));
  s := evalf(simpson(f,x=a..b,n));

                                n := 8
                                r := 0.7589932462
                                m := 0.8822686992
                                t := 0.8817037914
                                s := 0.8820655104

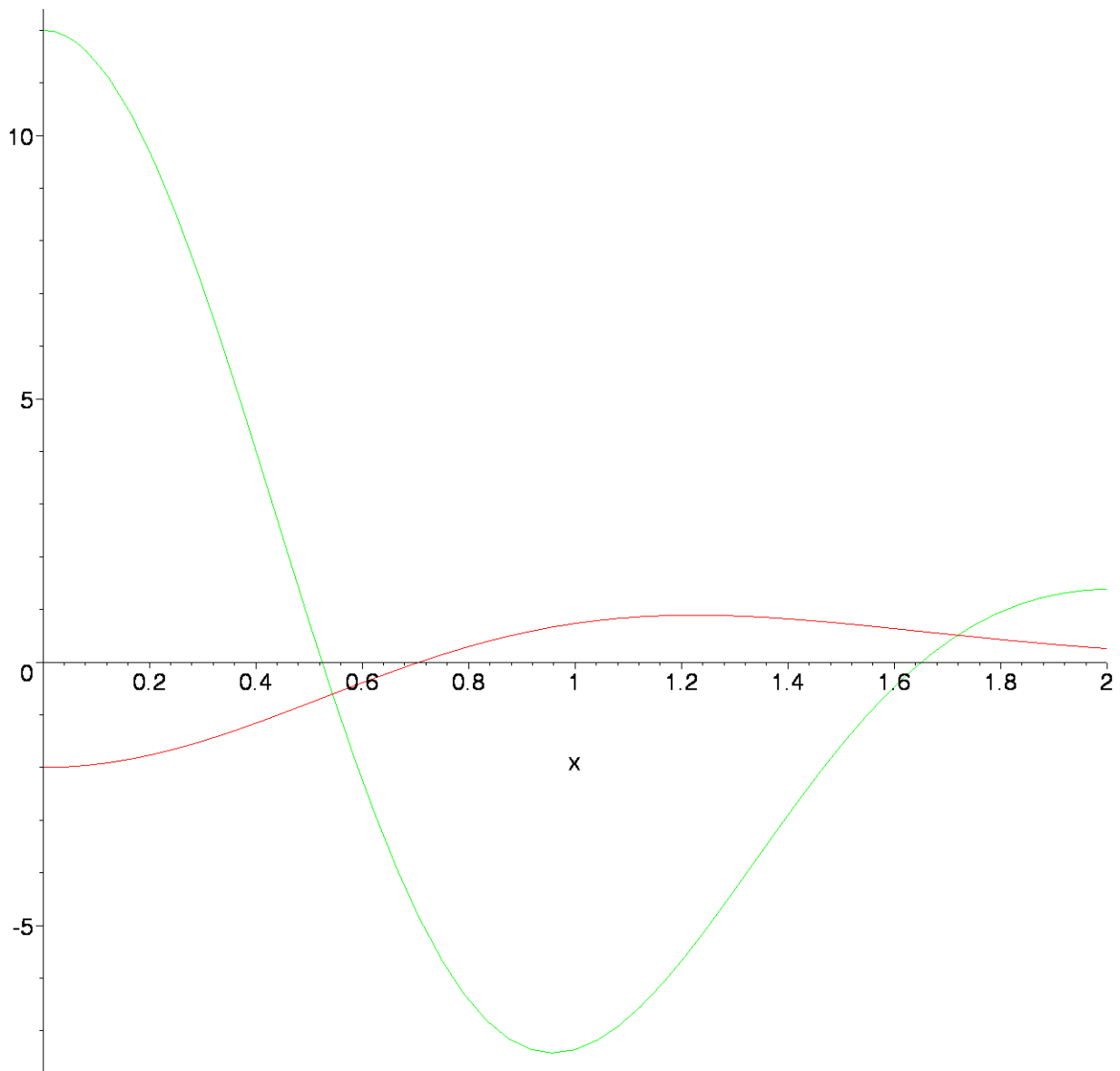
> fpp := diff(f,x$2);
  fpppp := diff(f,x$4);

```

```
plot([fpp, fpppp], x=a..b, color=[red, green]);
```

$$fpp := -2 e^{(-x^2)} + 4 x^2 e^{(-x^2)}$$

$$fpppp := 12 e^{(-x^2)} - 48 x^2 e^{(-x^2)} + 16 x^4 e^{(-x^2)}$$



```
> M := 2; K := 12;
err_mid := evalf((b-a)^3*(M)/(24*n^2));
err_trap := evalf((b-a)^3*(M)/(12*n^2));
err_simp := evalf((b-a)^5*(K)/(180*n^4));
```

```
M := 2
```

```
K := 12
```

```
err_mid := 0.01041666667
```

```
err_trap := 0.02083333333
```

```
| err_simp := 0.0005208333333
| > evalf(int(f,x=a..b));
| 0.8820813910
| >
```